THEME Evolution

Carboniferous Forests

Our industrial society runs on fossil fuels, such as coal. The term *fossil fuel* might seem odd at first until one realizes that it refers to the remains of organic material from ancient times. During the Carboniferous period, around 299 million years ago, a great swamp forest (Fig. 23A) encompassed what is now northern Europe, the Ukraine, and the Appalachian Mountains in the United States. The weather was warm and humid, and the plants grew very tall. These were not plants that would be familiar to us. Instead, they were related to various groups of plants that are less well-known—the lycophytes, horsetails, and ferns.

Lycophytes today may stand as high as 30 cm, but their ancient relatives were 35 m tall and 1 m wide. The strobili were up to 30 cm long, and some had leaves more than 1 m long. Horsetails, too—at 18 m tall—were giants compared to today's specimens. Tree ferns were also taller than the tree ferns found in the tropics today. The progymnosperms, including "seed ferns," so named because their leaves looked like fronds, were significant plants of a Carboniferous swamp.

The amount of biomass in a Carboniferous swamp forest was enormous, and occasionally the swampy water rose, covering the plants that had died. Dead plant material under water does not decompose well. The partially decayed remains became covered by sediment, which changed them into sedimentary rock. Exposed to significant amounts of pressure and over long periods of time, the organic material became coal. This process continued for millions of years, resulting in immense deposits of coal. Geological upheavals raised the deposits to the levels where they can be mined today.

With a change of climate, many of the plants of the Carboniferous period became extinct. Some of their smaller herbaceous relatives have evolved and survived to our time. We owe the industrialization of today's society to these ancient forests.

Questions to Consider

1. How might a geologist use this information to look for new sources of coal?
2. Why is coal not considered to be a renewable resource?



a.



b.

Figure 23A Swamp forest of the Carboniferous period. **a.** Nonvascular plants, early vascular plants, and early gymnosperms (often called "progymnosperms") dominated the swamp forests of the Carboniferous period. **b.** Micrograph of an early gymnosperm called a seed fern. (a): Richard Bizley/Science Source; (b): Sinclair Stammers/Science Source

modern trees. Today, horsetails have a rhizome that produces hollow, ribbed aerial stems and reach a height of 1.3 m (Fig. 23.13).

Equisetum can be branched or unbranched, with leaves that may have been megaphylls at one time but now are reduced and form whorls at the nodes. The skirtlike, slender, green side branches make the plant bear a resemblance to a horse's tail. Many horsetails have strobili at the tips of all stems; others send up buff-colored stems that bear the strobili. The spores germinate into inconspicuous and independent gametophytes. *Equisetum* spores are sensitive to humidity and are "spring-loaded." When conditions are right, the spores have been known to be ejected and travel upward of 2 miles in an hour.

The stems are tough and rigid because of silica deposited in cell walls. Early Americans, in particular, used horsetails for scouring pots and called them "scouring rushes." Today, they are still used as ingredients in a few abrasive powders.

Whisk Ferns

Whisk ferns are represented by the genera *Psilotum* and *Tmesipteris,* which are native to tropical and subtropical regions. The two *Psilotum* species resemble a whisk broom (Fig. 23.14), because they have no leaves. A horizontal rhizome gives rise to aboveground stems that repeatedly fork. The pumpkin-shaped sporangia are borne

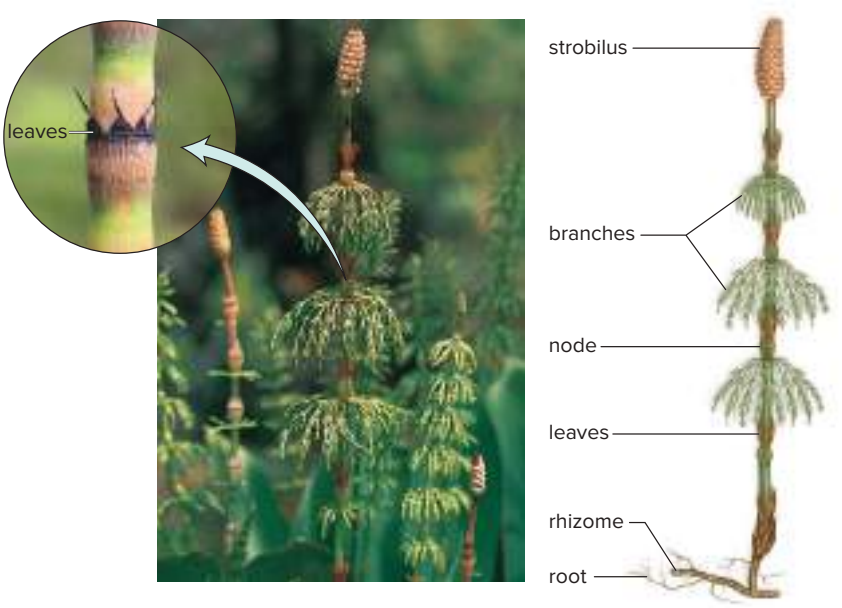


Figure 23.13 Horsetail, *Equisetum*. Whorls of branches and tiny leaves are at the nodes of the stem. Spore-producing sporangia are borne in strobili. (leaves): blickwinkel/Alamy Stock Photo; (plant): Robert Carr/Avalon

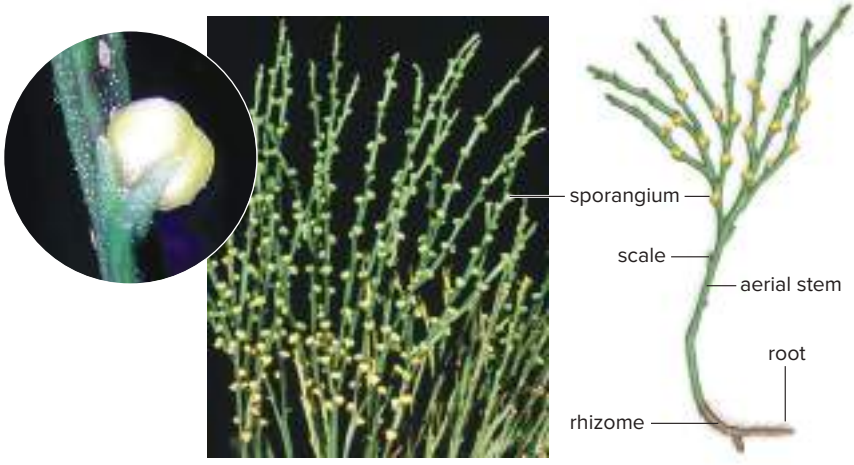


Figure 23.14 Whisk fern, *Psilotum*. *Psilotum* has no leaves—the branches carry on photosynthesis. The sporangia are yellow. (sporangium): Steven P. Lynch; (plant): Biophoto Associates/Science Source

on short side branches. The two or three species of *Tmesipteris* have appendages that some maintain are reduced megaphylls.

Ferns

Ferns are a widespread group of plants (~11,000 species) that are well-known for their attractiveness. Ferns are most abundant in warm, moist, tropical regions, but they can also be found in temperate regions and as far north as the Arctic Circle. Several species live in dry, rocky places, whereas others have adapted to an aquatic life. Ferns range in size from tiny aquatic species less than 1 cm in diameter to modern giant tropical tree ferns that exceed 20 m in height. Recall that unlike lycophytes, ferns have megaphylls, which are more commonly referred to as **fronds.** The leatherleaf fern (used in flower arrangements) has fronds that are broad, with subdivided leaflets; those of a tree fern can be about 1.4 meters long; and those of the hart's tongue fern are straplike and leathery (Fig. 23.15). Sporangia are often located in clusters, called *sori* (*sing., sorus*), on the undersides of the fronds, where they may be shielded by thin, protective structures called *indusia* (*sing., indusium*).

The life cycle of a typical temperate-zone fern, shown in Figure 23.16, applies in general to the other types of vascular seedless plants. The dominant sporophyte produces windblown spores by meiosis within sporangia. The windblown spores disperse, land, and germinate into the gametophyte, the generation that lacks vascular tissue. The separate, heart-shaped gametophyte produces flagellated sperm, which swim in a film of water from the antheridium to the egg within the archegonium, where fertilization occurs. Eventually, the gametophyte disappears and the sporophyte is independent. Both generations of a fern are considered to be independent of one another. Once established, some ferns, such as the bracken fern, can spread into drier areas, because their rhizomes, which grow horizontally in the soil, produce new plants.

Ferns have much economic value and are frequently used by florists in decorative bouquets and as ornamental plants in the home and garden. Wood from tropical tree ferns is often used as a building material, because it resists decay, particularly by termites. Ferns, especially the ostrich fern, are used as food—in the northeastern United States, many restaurants feature fiddleheads (that season's first



Figure 23.15 Diversity of ferns. Structural adaptations found in a variety of ferns. (leatherleaf fern): Gregory Preest/Alamy Stock Photo; (tree fern): ATOM WANG/Shutterstock; (hart's tongue fern): Nic Miller/Organics image library/Alamy Stock Photo

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